

Special Relativity Notes

Tensors: Lorentz Vectors, Covariant/Contravariant Components and Tensors of Rank n

5/10/26

Recap: Lorentz Transformation in Minkowski Space

Lorentz vector A^μ , $\mu = 0, 1, 2, 3$ (number of dimensions is 4).

Space and time components are not distinguishable:

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}$$

$$A^\mu = g^{\mu\nu} A_\nu, \quad A_\mu = g_{\mu\nu} A^\nu$$

Lorentz scalar: $A^\mu A_\mu = A^0 A_0 + A^i A_i$

$$A^\mu \Rightarrow A'^\mu = \Lambda^\mu_\alpha A^\alpha \quad \left(\Lambda^\mu_\alpha : \text{transformation matrix} \right)$$

Tensor

A **tensor** is a mathematical entity having n indices in d dimensions, so that the number of components is d^n .

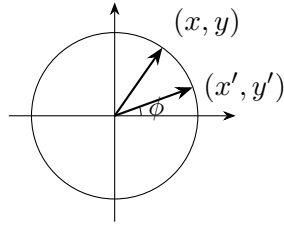
Ex: $A^{\mu\nu\sigma}$, $\mu = 0, 1, 2, 3$ (n -rank), $\nu = 0, 1, 2, 3$, $\sigma = 0, 1, 2, 3$
number of components $= 4^3 = 64$.

Ex: $F^{\mu\nu}$, $d = 4$, number of components $= 4^2 = 16$:

$$F^{\mu\nu} = \begin{pmatrix} f^{00} & f^{01} & f^{02} & f^{03} \\ f^{10} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix}_{4 \times 4}$$

Rotation Example

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$
$$x' = \cos \phi x + \sin \phi y, \quad y' = -\sin \phi x + \cos \phi y$$



$$\vec{X}' = R \vec{X}, \quad x'_i = R_{ij} x_j$$

$$\cos \phi = \frac{\partial x'}{\partial x} = \frac{\partial x'_1}{\partial x_1}, \quad \sin \phi = \frac{\partial x'}{\partial y} = \frac{\partial x'_1}{\partial x_2} \quad (x = x_1, y = x_2)$$

Tensor of rank $n = 0 \Rightarrow$ **scalar** Tensor of rank $n = 1 \Rightarrow$ **vector**

$$A = \sum_j a_j \hat{e}_j \quad (\text{in 3D}) \quad = a_1 \hat{e}_1 + a_2 \hat{e}_2 + a_3 \hat{e}_3$$

$$x'_1 = \frac{\partial x'_1}{\partial x_1} x_1 + \frac{\partial x'_1}{\partial x_2} x_2$$

$$x'_i = \sum_{j=1}^2 \frac{\partial x'_i}{\partial x_j} x_j$$

Contravariant and Covariant Vectors

$$A = \sum_{i=1}^n A_i \hat{e}_i = \sum_{i=1}^n A^i \hat{e}^i$$

Contravariant vector

$$A^i \Rightarrow A'^i = \sum_j \frac{\partial x'^i}{\partial x^j} A^j$$

Covariant vector

$$A_i \Rightarrow A'_i = \sum_j \frac{\partial x^j}{\partial x'^i} A_j$$

Tensors of Rank 2

A tensor of rank 2 can be written as

$$\underbrace{A^{\mu\nu}}_{\text{contravariant tensor}}, \quad \underbrace{A_{\mu\nu}}_{\text{covariant tensor}}, \quad \underbrace{A^\mu{}_\nu}_{\text{mixed tensor}}$$

$$A^{\mu\nu} \longrightarrow A'^{\mu\nu} = \sum_{\alpha,\beta} \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x'^{\nu}}{\partial x^{\beta}} A^{\alpha\beta}$$

$$A_{\mu\nu} \longrightarrow A'_{\mu\nu} = \sum_{\alpha,\beta} \frac{\partial x^{\alpha}}{\partial x'^{\mu}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} A_{\alpha\beta}$$

$$A^{\mu}{}_{\nu} \longrightarrow A'^{\mu}{}_{\nu} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} A^{\alpha}{}_{\beta}$$

Ex:

$$A^{\mu\nu\sigma} \longrightarrow A'^{\mu\nu\sigma} = \frac{\partial x'^{\mu}}{\partial x^{\alpha}} \frac{\partial x'^{\nu}}{\partial x^{\beta}} \frac{\partial x'^{\sigma}}{\partial x^{\gamma}} A^{\alpha\beta\gamma}$$